Anaïs Azouaoui

MSc student in Computer Science, seeking a 6-month research internship from March 2027

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Research interests: reliable AI, computer vision & LLMs/VLMs, robot & reinforcement learning, evolutionary computation

Education

Master's in Computer Science, Sorbonne University, Paris — *2025 – 2027 (expected)*
Algorithms, AI, Interactions & Decision (AI2D)

BSc in Computer Science, Sorbonne University, Paris — *2022 – 2025*

Research Experience

Real-to-Sim Scene Generation | *Research Project* — Feb – May 2026
ISIR, Sorbonne University, Paris — Supervisors: S. Doncieux, E. Garrabe

- Built a multi-step **LLM/VLM** pipeline turning a prompt or a photo into a physically valid **Genesis** scene.
- Generated training data by paraphrasing sampled JSON scene targets, filtered by an **LLM verifier**.
- **Fine-tuned** the **Phi-3-mini** LLM with **QLoRA** in **PyTorch** on 3,531 synthetic examples to run fully locally and improved spatial-relation F1 from 56% (GPT-4o pipeline) to 72% at comparable latency (∼8 s).
- Reduced position error from 1.25 to 0.48 m on real Franka Panda scenes with **GroundingDINO**.
- Matched free-form object labels to a 3D asset catalog by **embedding retrieval** (**multilingual-e5**).
- Built a **VLM verification loop** with **Genesis** physics checks, restricted to symbolic corrections.

Interpretable ML for Intensive Care | *Research Internship* — Jun – Aug 2026
IRIT, REVA team - CHU de Toulouse, Toulouse — Supervisors: S. Cussat-Blanc, D. Wilson, G. Nadizar

- Modeled clinician-rated severity of 4 respiratory disorders in 164 ventilated ICU patients.
- Benchmarked 16 ML models (linear, kernel, trees, MLP); the best reached 82–96% of the inter-rater ceiling.
- Built a **Cartesian Genetic Programming** engine with **NSGA-II** to find interpretable formulas; formulas with 2–3 variables reach 94–111% of the best ML model's R^2 (Extra Trees).
- Tested 5 **LLM-assisted formula search** methods; the best beat Extra Trees on one disorder (R^2 0.86 vs. 0.81).

Projects

Damage Recovery for a Quadruped Robot — 2026 | Project page

- Reproduced **Intelligent Trial & Error** (ITE, Cully et al., Nature 2015) on a simulated quadruped.
- Built gait repertoires in **JAX**; after damage, **PGA-ME/DCRL-ME** gaits ran 5–13× faster than **MAP-Elites** ones.
- Adapted to 12 leg-damage scenarios with **ITE** in a median of 5 trials, nearly doubling speed vs. no adaptation.
- Benchmarked ITE against **TD3** fine-tuning, stabilized with a critic warm-up: ITE needed 90× fewer episodes.

Vision-Language-Action Model Robustness — 2026 | Project page

- Stress-tested **SmolVLA** on 3,688 **LIBERO** episodes (camera, light, arm); a 3° per-joint shift halves success.
- Ran **metamorphic tests** on instructions: rewording them without changing their meaning dropped success from 80% to 23%; **embedding retrieval** restored 80%.
- Used **MAP-Elites** to find mixes of small perturbations, each harmless alone, that fail 71% of held-out episodes.

Sign Language Recognition — 2026 | Project page

- Trained a **Conv-Transformer** on **MediaPipe** landmarks, beating a linear baseline by 20 points on unseen signers.
- Ran 10 **one-factor ablations**, identifying dominant-hand canonicalization as the key step (+3.8 points).

Skills

Domains	Deep Learning, Reinforcement Learning, Computer Vision, LLMs/VLMs, NLP, Robot Learning
Methods	Transformers, Fine-Tuning (QLoRA), Metaheuristic Optimization (Genetic Programming, NSGA-II, MAP-Elites), Bayesian Optimization
Tools	Python, C++, PyTorch, JAX, Hugging Face, scikit-learn, Simulation (Genesis, LeRobot), Git, Docker
Languages	French (native), English (proficient), Arabic (proficient), Japanese (basic)

Manuscripts in Preparation

[1] **A. Azouaoui**, C. Zouloumian, E. Garrabe, S. Doncieux. *Real-to-Sim: generating physically valid robot simulation scenes from text and images with LLMs and VLMs.*

[2] **A. Azouaoui**, G. Nadizar, S. Cussat-Blanc, D. Wilson, M. Poette. *Interpretable prediction of clinician-rated respiratory severity in intensive care with genetic programming.*